Task 1:

```
bodike@bodike-chaithali:~/Home/Lab3$ ls
bodike@bodike-chaithali:~/Home/Lab3$ mkdir activity
bodike@bodike-chaithali:~/Home/Lab3$ ls
activity
bodike@bodike-chaithali:~/Home/Lab3$ cd activity/
bodike@bodike-chaithali:~/Home/Lab3/activity$ ls
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat > emp_data.txt << EOF
> 101,Anjali Sharma,HR,Manager,850000
> 102,Ravi Verma,IT,SoftwareEngineer,720000
> 103,Priya Desai,Marketing,Executive,600000
> 104,Amitabh Roy,Finance,Analyst,780000
> 105,Sneha Iyer,IT,QA Tester,550000
> 106,Karan Mehta,HR,Recruiter,575000
> 107,Neha Kulkarni,IT,DevOps Engineer,820000
> 108,Arjun Singh,IT,Manager,900000
> 109,Ritu Gupta,Finance,Accountant,650000
> 110,Suresh Patil,IT,Tech Lead,1000000
> EOF
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat emp_data.txt
101,Anjali Sharma,HR,Manager,850000
102,Ravi Verma,IT,SoftwareEngineer,720000
103,Priya Desai,Marketing,Executive,600000
104,Amitabh Roy,Finance,Analyst,780000
105,Sneha Iyer,IT,QA Tester,550000
106,Karan Mehta,HR,Recruiter,575000
107,Neha Kulkarni,IT,DevOps Engineer,820000
108,Arjun Singh,IT,Manager,900000
109,Ritu Gupta,Finance,Accountant,650000
110,Suresh Patil,IT,Tech Lead,1000000
```

Q1)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat emp_data.txt | sort -t',' -k5 -nr | awk -F',' '$5 > 600000' | sed 's/,/:/g' > senior_emp.txt

bodike@bodike-chaithali:~/Home/Lab3/activity$ cat senior_emp.txt
110:Suresh Patil:IT:Tech Lead:1000000
108:Arjun Singh:IT:Manager:900000
101:Anjali Sharma:HR:Manager:850000
107:Neha Kulkarni:IT:DevOps Engineer:820000
104:Amitabh Roy:Finance:Analyst:780000
102:Ravi Verma:IT:SoftwareEngineer:720000
109:Ritu Gupta:Finance:Accountant:650000
bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Q2)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat senior_emp.txt
110:Suresh Patil:IT:Tech Lead:1000000
108:Arjun Singh:IT:Manager:900000
101:Anjali Sharma:HR:Manager:850000
107:Neha Kulkarni:IT:DevOps Engineer:820000
104:Amitabh Roy:Finance:Analyst:780000
102:Ravi Verma:IT:SoftwareEngineer:720000
109:Ritu Gupta:Finance:Accountant:650000
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat senior_emp.txt | sort -t':' -k1 -n | awk -F':' '$3=="IT" {count++; if(count > 2) print $0}'
108:Arjun Singh:IT:Manager:900000
110:Suresh Patil:IT:Tech Lead:1000000
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat senior_emp.txt
110:Suresh Patil:IT:Tech Lead:1000000
108:Arjun Singh:IT:Manager:900000
101:Anjali Sharma:HR:Manager:850000
107:Neha Kulkarni:IT:DevOps Engineer:820000
104:Amitabh Roy:Finance:Analyst:780000
102:Ravi Verma:IT:SoftwareEngineer:720000
109:Ritu Gupta:Finance:Accountant:650000
bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Q3)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat emp_data.txt | awk -F',' '$2~/A/' | sort -t',' -k5 -nr |head -1 |awk -F',' '{print $1"::"$2"
::"$4}'
108::Arjun Singh::Manager
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat emp_data.txt
101,Anjali Sharma,HR,Manager,850000
102,Ravi Verma,IT,SoftwareEngineer,720000
103,Priya Desai,Marketing,Executive,600000
104,Amitabh Roy,Finance,Analyst,780000
105,Sneha Iyer,IT,QA Tester,550000
106,Karan Mehta,HR,Recruiter,575000
107,Neha Kulkarni,IT,DevOps Engineer,820000
108,Arjun Singh,IT,Manager,900000
109,Ritu Gupta,Finance,Accountant,650000
110,Suresh Patil,IT,Tech Lead,1000000
bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Q4)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ grep -v "$(cat emp_data.txt | awk -F',' '$3=="IT" ' |sort -t',' -k5 -nr | sed -n '2p')" emp_data
.txt > temp && mv temp emp_data.txt
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat emp_data.txt
101,Anjali Sharma,HR,Manager,850000
102,Ravi Verma,IT,SoftwareEngineer,720000
103,Priya Desai,Marketing,Executive,600000
104,Amitabh Roy,Finance,Analyst,780000
105,Sneha Iyer,IT,QA Tester,550000
106,Karan Mehta,HR,Recruiter,575000
107,Neha Kulkarni,IT,DevOps Engineer,820000
109,Ritu Gupta,Finance,Accountant,650000
110,Suresh Patil,IT,Tech Lead,1000000
bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Task 2:

Q1)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp1s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 28:00:af:b4:42:36 brd ff:ff:ff:ff:ff:ff
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether e8:65:38:0e:9d:a7 brd ff:ff:ff:ff:ff:ff
    inet 10.50.33.181/18 brd 10.50.63.255 scope global dynamic noprefixroute wlp2s0
       valid_lft 14395sec preferred_lft 14395sec
    inet6 fe80::468:9317:373e:1cf7/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Q2)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ ping -c 4 iitbhilai.ac.in
PING iitbhilai.ac.in (192.168.10.115) 56(84) bytes of data.
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=1 ttl=62 time=1.93 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=2 ttl=62 time=6.87 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=3 ttl=62 time=2.28 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=4 ttl=62 time=6.30 ms

--- iitbhilai.ac.in ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.932/4.345/6.868/2.251 ms
bodike@bodike-chaithali:~/Home/Lab3/activity$ nslookup iitbhilai.ac.in
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   iitbhilai.ac.in
Address: 192.168.10.115

bodike@bodike-chaithali:~/Home/Lab3/activity$
```

Q3)

```
bodike@bodike-chaithali:~/Home/Lab3/activity$ vi subString_count.c
bodike@bodike-chaithali:~/Home/Lab3/activity$ cat subString_count.c
#include<stdio.h>
#include<string.h>

int main(int argc,char *argv[]) {
        if(argc!=3) {
                printf("Usage: %s <substring> <filename>\n",argv[0]);
                return 1;
        }
        char *subString=argv[1];
        char *filename=argv[2];
        FILE *file=fopen(filename,"r");

        if(file==NULL) {
                printf("Error in opening file %s\n",filename);
                return 1;
        }
        char line[1000];
        int count =0;
        int sub_len=strlen(subString);
        while(fgets(line,sizeof(line),file)) {
                for(int i=0;line[i]!='\0';i++) {
                        if(strncmp(&line[i],subString,sub_len)==0) {
                                count++;
                        }
                }
        }

        fclose(file);
        printf("%d %s\n",count,filename);
        return 0;
}
bodike@bodike-chaithali:~/Home/Lab3/activity$ gcc subString_count.c -o subString_count
bodike@bodike-chaithali:~/Home/Lab3/activity$ ./subString_count  IT  emp_data.txt
4 emp_data.txt
bodike@bodike-chaithali:~/Home/Lab3/activity$
```